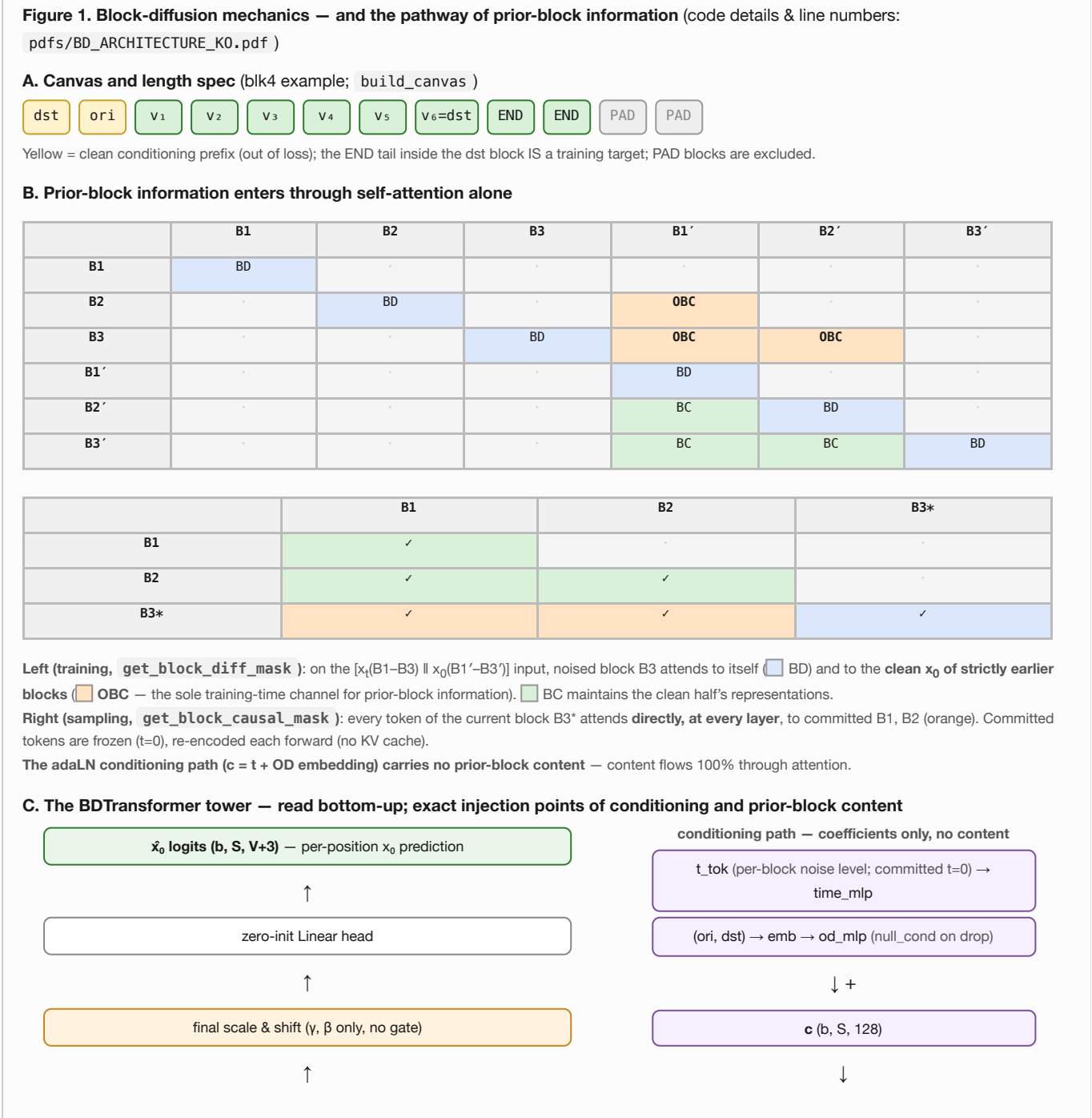


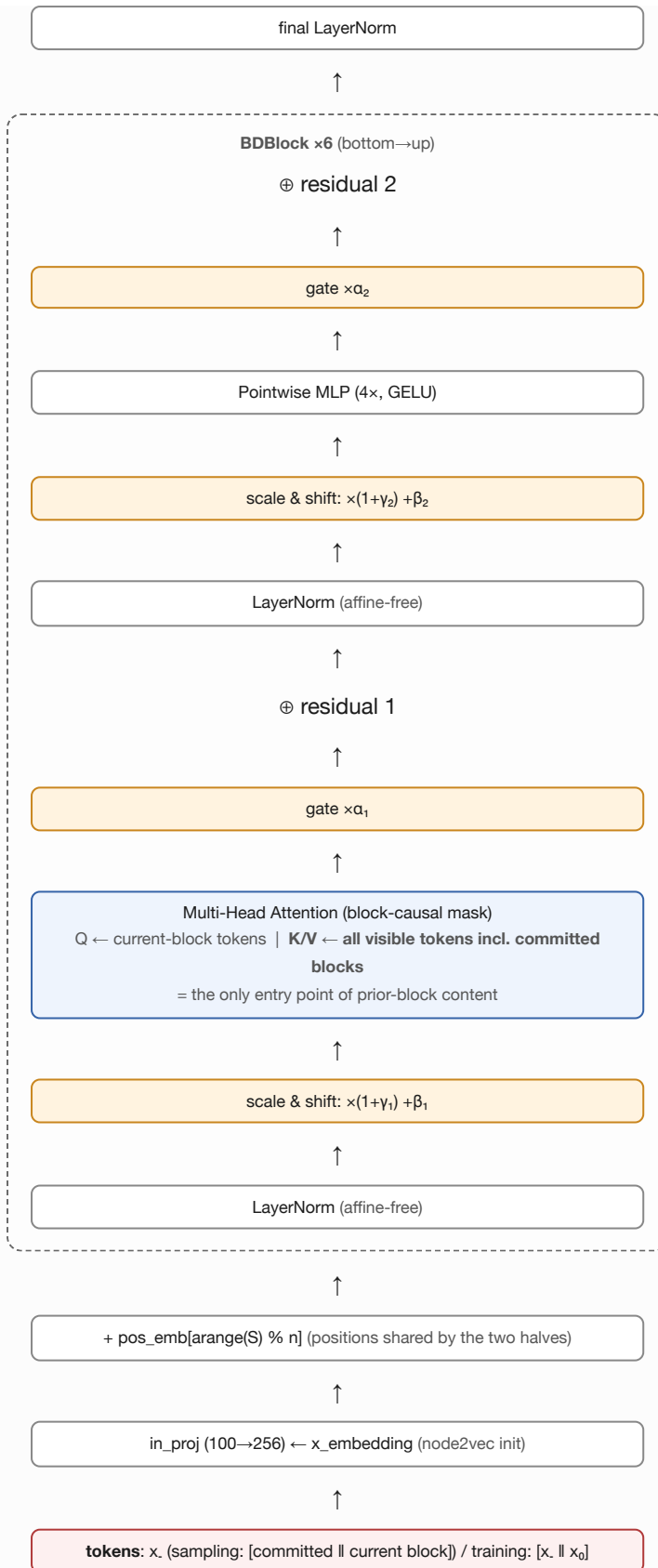
Block-Diffusion OD Planning — Consolidated Results (v3)

Status (2026-07-20, v3). Every number in this document is re-measured on the **v2 backbones** (14 models retrained with seeded 80/20 splits: mask×7, graph d=2.0×7). Evaluation: the normal task uses a **shared held-out 1,000-pair set** drawn from the untouched test split (§2: 3 generation seeds, mean±std); the exceptional task uses the except_0 **reserved 1,000 pairs** excluded from discriminator training. The developmental narrative and full ablation history live in RESULTS_BD_v1.pdf (archived); full derivations in BD_GUIDANCE_FORMULATION.pdf , BD_CODE_VERIFICATION_K0/EN.pdf , BD_ARCHITECTURE_K0.pdf . v1 checkpoints/discs archived as v1_* .

1. Setup and architecture

BD3-LM-style block diffusion (semi-AR across blocks, discrete diffusion within), 6-layer / dim-256 DiT (adaLN), porto_v3_normal (1,390 vertices, 1.93M paths), 1 epoch = 48.2k iters (bs 32, lr 5e-4). Canvas [dst, ori, v..., v_L=dst, END... | PAD...] : the END tail inside the dst block is supervised, PAD blocks excluded. Kernels: mask (absorbing, SUBS, first-hitting — max data length 59, so blk64 is effectively MDLM) and graph (uniform-state CTMC, d = 2.0, T = 100). Generation is open-ended (dst→hit / END→miss).





adaLN MLP (Mish→Linear, **zero-init**)

→ $(\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2)$

adaLN feeds every $\gamma/\beta/\alpha$ box in every block

↖ all orange boxes

- γ, β, α are **NOT** model parameters. The learned parameters are the purple adaLN MLP's weights (zero-init → every block starts as identity); $\gamma/\beta/\alpha$ are activations computed from c per token and per sample. Conditioning never touches weights — it scales $(1+\gamma)$ /shifts (β) the normalized activations feature-wise and gates (α) sublayer outputs (why the LayerNorms are affine-free).
- Prior-block content enters through the attention K/V (blue box), not through adaLN. In short: **content via K/V ; conditioning (per-block t , OD) via $\gamma\beta\alpha$.**

```
( $\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2$ ) = adaLN_MLP(c).chunk(6) # BDBlock.forward, bd_models.py:149-153
x = x +  $\alpha_1 \odot \text{Attn}((1+\gamma_1)\odot\text{LN}(x) + \beta_1, \text{attn\_mask})$ 
x = x +  $\alpha_2 \odot \text{MLP}((1+\gamma_2)\odot\text{LN}(x) + \beta_2)$ 
```

D. Open-ended semi-AR generation: append block → first-hitting reveals one token per forward (each revealed token immediately becomes context) → scan the committed block: dst → hit (stop) / END → miss (stop) / else append the next noised block.

Training's OBC (noised → clean past) corresponds exactly to sampling's committed-block reference (teacher forcing), so the conditioning distributions match. At blk1 all information flows through committed references (a causal LM); at blk64 (= MDLM) OBC is empty — block size interpolates the ratio of the two channels (§6.9).

2. Block-size sweep — held-out, 3 seeds mean±std

2.1 mask

blk	hit (=arr)	valid	valid∧hit (raw)	EM	PC	DTW	len err	len corr
1	0.997±.002	0.934±.007	0.934±.007	0.830±.011	0.869±.010	0.032	0.84	+0.864
2	0.958±.007	0.842±.006	0.822±.006	0.769±.005	0.799±.005	0.038	0.45	+0.982
4	0.958±.001	0.836±.004	0.811±.004	0.747±.010	0.783±.008	0.037	0.37	+0.991
8	0.951±.002	0.808±.006	0.774±.006	0.716±.008	0.754±.007	0.042	0.40	+0.993
16	0.898±.006	0.680±.006	0.625±.002	0.607±.009	0.637±.008	0.061	0.56	+0.989
32	0.858±.010	0.474±.004	0.420±.002	0.413±.002	0.437±.002	0.081	0.96	+0.968
64	0.839±.005	0.430±.014	0.373±.010	0.366±.012	0.391±.012	0.072	0.89	+0.988

2.2 graph (d = 2.0)

blk	hit (=arr)	valid	valid∧hit (raw)	EM	PC	DTW	len err	len corr
1	0.429±.001	1.000±.000	0.429±.001	0.093±.006	0.168±.007	0.437	36.4	+0.237
2	0.173±.014	0.997±.001	0.173±.014	0.399±.008	0.558±.005	0.488	6.48	+0.492
4	0.128±.007	0.978±.002	0.128±.007	0.309±.012	0.499±.008	0.302	4.50	+0.719
8	0.058±.004	0.991±.001	0.058±.004	0.124±.012	0.331±.006	0.282	4.53	+0.783
16	0.029±.002	0.996±.002	0.029±.002	0.148±.008	0.390±.006	0.377	5.66	+0.705
32	0.035±.002	0.969±.002	0.035±.002	0.174±.002	0.411±.004	0.394	5.96	+0.693
64	0.027±.004	0.966±.003	0.027±.004	0.198±.006	0.430±.001	0.363	5.74	+0.724

2.3 Failure anatomy (v3, 400 held-out pairs)

blk	hit	len corr (hit)	len corr (miss, n)	miss→dst med hops	≤2 hops	inv-edge	boundary/interior
4	0.943	0.996	0.958 (23)	2	87%	1.04%	15 / 77
16	0.922	0.995	0.896 (31)	2	84%	1.84%	4 / 158
64	0.843	0.993	0.968 (63)	1	86%	4.70%	0 / 409

- **A miss is a near-miss:** endpoints sit a median 1–2 hops from dst (84–87% ≤2) — a last-token identification slip, not a navigation failure. Length calibration holds for hits (0.99+) and misses (0.90–0.97) — dst-truncation geometry plus END calibration.
- **The validity decay is path-level amplification of sparse local skips:** the edge error rate is only 4.7% even at blk64 (median 3-hop skips), and the errors are **strictly block-interior** (blk64: 0 boundary / 409 interior) — first-hitting's random within-block reveal order is the mechanism; block-boundary (AR-like) transitions are clean.
- **The graph kernel is the mirror image:** constrained decoding keeps validity at 0.97–1.00 but arrival collapses 0.43→0.03 and decays with length; mask shows no horizon decay for blk1–16. Block size charges mask in validity, graph in arrival.
- **Held-out validation:** v2 numbers match v1 (train-overlapping) within noise — no memorization inflation.
- **Untested note:** the END tail's loss share grows as (B–1)/2 — ~58% of supervised tokens at blk64 — leaving a loss-engineering lever (constant-mass reweighting / first-END-only; v1 §4.5).

3. Post-processing: P1 (local splice) · P3 (endpoint patch)

P1 inserts a shortest-path bridge at every invalid pair (u,v) → valid = 1 by construction, hit untouched. P3 (= refine_nx) grafts a shortest path from a miss endpoint to dst → arrival = 1. hit is always measured pre-refinement (valid∧hit (raw)) — the one model-dependent quantity left under P1+P3.

blk	raw arr	raw valid	valid∧hit (raw)	raw EM	raw PC	P1+P3 EM	P1+P3 PC	patch
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1	0.997	0.934	0.934	0.832	0.871	0.849	0.898	1.0
2	0.956	0.852	0.826	0.779	0.810	0.804	0.869	0.5
4	0.954	0.823	0.798	0.734	0.771	0.768	0.846	0.8
8	0.954	0.827	0.794	0.728	0.769	0.768	0.848	0.7
16	0.909	0.705	0.647	0.617	0.651	0.662	0.766	1.9
32	0.871	0.462	0.408	0.397	0.422	0.503	0.642	3.3
64	0.853	0.441	0.390	0.379	0.403	0.533	0.661	3.5

- P1+P3 gives the structural **valid = arrival = 1** guarantee at every block size, with usable-path rate = raw hit. At large blocks P1 also lifts EM/PC (blk64 EM 0.379→0.533) — the bridges are real route segments. Zero unfixable pairs.

4. Importance-weight guidance (exceptional scenario except_0)

Math summary (full: BD_GUIDANCE_FORMULATION.pdf). Guided reverse of block b: $q(x_{t-1}^b | x_t^b, x^{<b}) = \mathbb{E}_{x_0 \sim p}[w \cdot q(\cdot | x_0)] / \mathbb{E}[w]$, $w = q(x_0 | x^{<b}) / p_\theta(x_0 | x^{<b}) \approx D / (1 - D)$. **Lemma 1**: $w = C^{-1} \cdot r(x^{<b} \| x_0)$ with $C = q(x^{<b}) / p_\theta(x^{<b})$ candidate-independent, cancelling under self-normalization — a **clean partial-path** discriminator suffices. **Lemma 2**: a block-only classifier learns the marginal ratio and is biased (non-constant mismatch). **Lemma 3**: masking the proposal to A_{scn} -legal transitions w.r.t. revealed neighbours is exact — excluded candidates lie outside $\text{supp}(q)$ ($w = 0$) and Z_ℓ is candidate-common, cancelling (unbiased variance reduction). Implementation: $N = 100$ mean-field candidates per first-hitting reveal, $w = \exp(\text{bounded logit})$, applied at every reveal, ESS logged. Discriminator: GraphSAGE + edge-existence channels (scenario adjacency as input → zero-retraining scenario transfer); negatives = **20k unconditional generations of the very block's model** (the true denominator p_θ ; the data<mix>model ordering: v1 §6.7).

4.1 Seen (disc trained on except_0, 1%)

blk	base valid	base EM	IW valid	IW EM	IW P1P3 EM	IW P1P3 PC	ESS	adj+IW valid	adj+IW EM	rem-edge	adj+IW P1P3 EM	adj+IW P1P3 PC
4	0.266	0.240	0.331	0.298	0.309	0.464	99.1	0.390	0.338	3.0%	0.357	0.496
8	0.231	0.212	0.298	0.264	0.294	0.453	98.8	0.346	0.298	3.2%	0.335	0.483
16	0.222	0.198	0.277	0.240	0.268	0.423	98.2	0.327	0.270	3.1%	0.295	0.447
32	0.160	0.132	0.225	0.184	0.216	0.376	97.5	0.255	0.203	3.2%	0.236	0.398
64	0.141	0.129	0.201	0.160	0.211	0.372	95.4	0.229	0.176	3.1%	0.230	0.394

4.2 Unseen (disc trained on except 1–99 → evaluated on except_0)

blk	IW valid (unseen/seen)	adj+IW valid (unseen/seen)	adj+IW P1P3 EM (unseen/seen)
4	0.325 / 0.331	0.382 / 0.390	0.348 / 0.357
16	0.270 / 0.277	0.321 / 0.327	0.299 / 0.295

- **Division of labour**: removed-edge avoidance comes from Lemma 3 masking (rem-edge ~5%→3%), route quality from discriminator reweighting (+5–7%p valid). ESS ≈ 95–99: per-step nudges are small but accumulate over reveals. The cost is arrival (–5 to –7%p) — restored structurally by P3.
- **Generalization gap ≈ 0** (blk4: 0.382 vs 0.390): scenario information enters through the adjacency input, not memorization. The full stack (adj+IW+P1P3) reaches EM 0.357 / PC 0.496 at blk4 — +32%/+15% relative over baseline+P1P3 (0.270/0.431).

5. Controlled comparison with the published baseline: D-CBG (Schiff et al.)

Plug-in port (their code logic verbatim; dcbg_plugin.py), **classifier architecture and training protocol fully matched** (same GraphSAGE + edge channels, same 1% data; model negatives use the same v2 uncond pools, noised their way). The only remaining differences are method-intrinsic: noisy-state input + per-position tilting (D-CBG) vs clean partial paths + joint candidate reweighting (IW). $\gamma = 4$ ($\gamma \leq 2$ inert, v1 §7.1). Exact enumeration costs $|V| \approx 1.4\text{k}$ classifier calls per reveal (≈2.7 s/path); the cheap first-order variant (0.07 s/path) is **re-confirmed inert on v2** (blk4 valid 0.265 ≈ base 0.266); IW: 2.4k calls / 0.03 s/path.

blk	D-CBG(data) valid	P1P3 EM	D-CBG(model) valid	P1P3 EM	IW(model) valid	P1P3 EM	adj+IW valid	P1P3 EM
4	0.301	0.295	0.309	0.302	0.331	0.309	0.390	0.357
8	0.257	0.275	0.272	0.275	0.298	0.294	0.346	0.335
16	0.249	0.263	0.258	0.252	0.277	0.268	0.327	0.295
32	0.185	0.231	0.185	0.203	0.225	0.216	0.255	0.236
64	0.163	0.222	0.173	0.209	0.201	0.211	0.229	0.230

- **IW > D-CBG at every block size** (raw valid 10/10; blk4 0.331 vs 0.309), and the full stack adj+IW tops every cell. Our ingredients — the scenario-conditioned classifier and model negatives — port to D-CBG and improve it, yet the ordering persists.
- **Unseen transfer holds for both, with the IW lead intact:** blk4 D-CBG 0.306 / IW 0.325 (seen 0.309 / 0.331). Generalization is a property of the adjacency-conditioned classifier design (our contribution, portable).
- **Cost-quality frontier:** the cheap D-CBG (first-order) does not work and the working D-CBG (exact) is not cheap — IW dominates the region between (30–90× cheaper, no γ to tune, arrival preserved). On the uniform kernel their noisy-state classifier is structurally untrainable (chance level), making the gap decisive (v1 §7).

6. Conclusions and artifacts

- Small mask blocks (1–8) essentially solve the task; the residual defects (local skips, near-misses) are structurally repaired by P1/P3. The graph kernel is the opposite pole of the validity–arrival trade.
- The full exceptional-scenario system — **adj-masked proposals + model-negative partial-path discriminator (IW) + P1/P3** — is best on seen and unseen scenarios and beats the published baseline (D-CBG) in every matched configuration.
- Models: sets_model/*_v2_bd.pth (14; v1 under v1_archive/). Results: sets_res/, logs: sets_log/v3_*. Code: bd_models.py (plan_guided), bd_disc.py, dcbg_plugin.py, train_bd_disc.py, train_dcbg_classifier.py, eval_*.py, run_v3_all.sh.